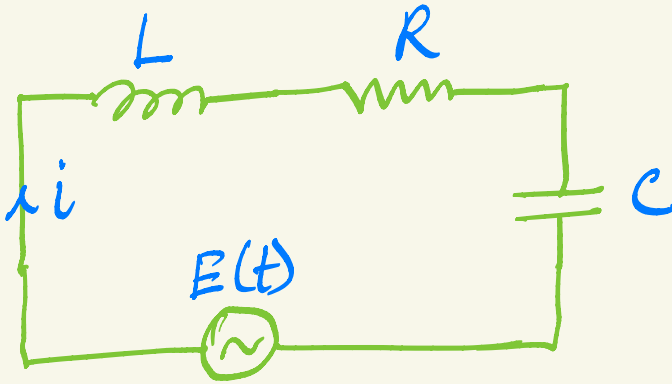




Lets consider the example of an electrical system.



In this circuit there are three elements connected in series: an inductor, a resistor and a capacitor.

L is called inductance

R is called Resistance

C is called capacitance

$E(t)$ is the applied source voltage.
It is a function of time.

Due to the application of the voltage, current i flows through the circuit.

The voltage drop across the inductor is

$$V_L = L \frac{di}{dt}$$

The voltage drop across the resistor is $V_R = iR$

The voltage drop across the capacitor is

$V_C = \frac{1}{C} \int i dt$, where i is the current.

Current i can be expressed in terms of the charge q

$$i = \frac{dq}{dt}$$

As a result, the voltage drop equations in terms of charge q is as follows

$$V_L = L \frac{dq}{dt}$$

$$V_R = R \frac{dq}{dt}$$

$$V_C = \frac{1}{C} q$$

The Kirchhoff's voltage law states that the sum of the voltage drop is equal to the source voltage.

$$V_L + V_R + V_C = E(t)$$

$$\Rightarrow L \frac{d^2 q(t)}{dt^2} + R \frac{dq(t)}{dt} + \frac{q(t)}{C} = E(t)$$

$$\Rightarrow \frac{d^2 q(t)}{dt^2} + \frac{R}{L} \frac{dq(t)}{dt} + \frac{1}{LC} q(t) = \frac{E(t)}{L} \quad (1)$$

Note that this equation has the same form as the spring mass damper system.

By analogy with a spring mass system,

charge $q(t) \rightarrow$ displacement $x(t)$

current $i(t) \rightarrow$ velocity $\dot{x}(t)$

voltage $e(t) \rightarrow$ force $f(t)$

inductance $L \rightarrow$ mass m

Resistance $R \rightarrow$ damping b

capacitance $\frac{1}{C} \rightarrow$ stiffness k

As a result, the spring mass system can be modeled with a circuit diagram.

Take Laplace transform of the differential equation (1)

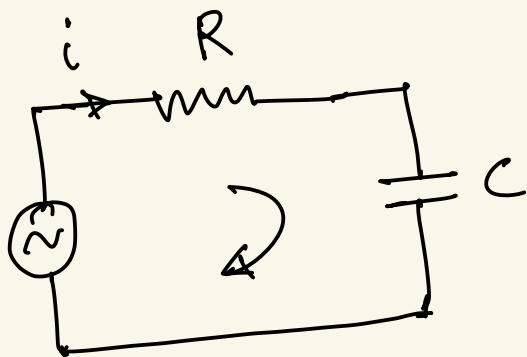
$$s^2 Q(s) + \frac{R}{L} s Q(s) + \frac{1}{LC} Q(s) = E(s)$$

$$\Rightarrow \frac{Q(s)}{E(s)} = G(s) = \frac{1}{s^2 + \frac{R}{L} s + \frac{1}{LC}}$$

$$\Rightarrow G(s) = \frac{LC}{LC s^2 + RC s + 1}$$

$G(s)$ is the transfer function of the system. This is a second order system.

Let's consider the below circuit which is a first order system.



$\Sigma \text{Voltage drop} = \Sigma \text{source voltage}$

$$\Rightarrow V_R + V_C = E(t)$$

$$\Rightarrow iR + \frac{1}{C} \int i dt = E(t)$$

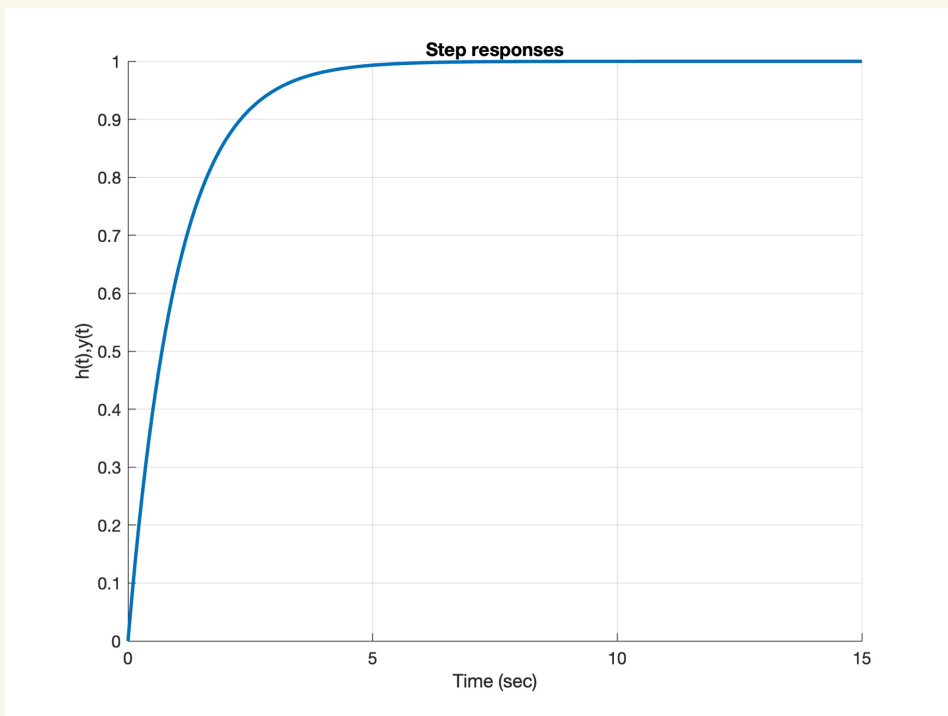
$\Rightarrow$ Take Laplace

$$\Rightarrow I(s)R + \frac{1}{Cs} I(s) = E(s)$$

$$\Rightarrow \frac{I(s)}{E(s)} = \frac{1}{R + \frac{1}{Cs}}$$

$$G(s) = \frac{Cs}{RCs + 1}$$

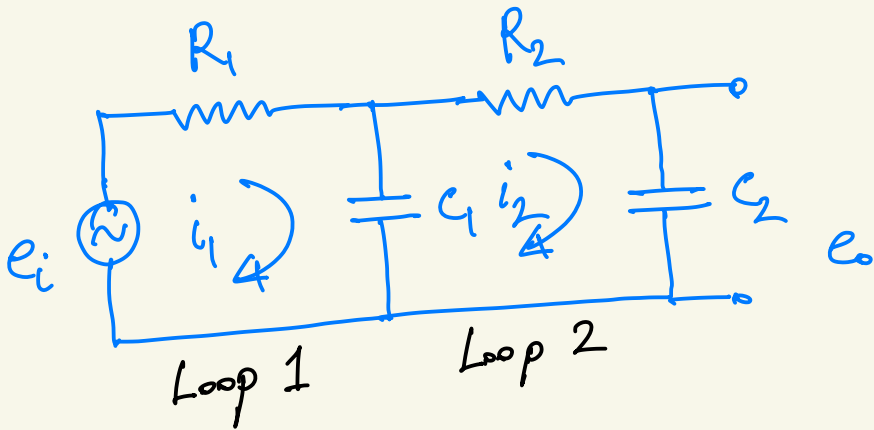
$RC = \tau = \text{time constant}$



This figure shows the step response of the transfer function

$G(s)$ when $RC = \gamma = 1$. Note that there is no overshoot in a first order system.

Let's solve and find the transfer function of a coupled system.



From loop 1 :

$$i_1 R_1 + \frac{1}{C_1} \int (i_1 - i_2) dt = e_i$$

From loop 2:

$$i_2 R_2 + \frac{1}{C_1} \int (i_2 - i_1) dt + \frac{1}{C_2} \int i_2 dt = 0$$

Take Laplace transform:

$$I_1(s) R_1 + \frac{1}{C_1} \cdot \frac{1}{s} [I_1(s) - I_2(s)] = E_i(s)$$

$$I_2(s) R_2 + \frac{1}{C_1} \cdot \frac{1}{s} [I_2(s) - I_1(s)] + \frac{1}{C_2} \cdot \frac{1}{s} I_2(s) = 0$$

Rearrange:

$$\left(R_1 + \frac{1}{C_1 s}\right) I_1(s) - \frac{1}{C_1 s} I_2(s) = E_i(s)$$

$$-\frac{1}{C_1 s} I_1(s) + \left(\frac{1}{C_1 s} + R_2 + \frac{1}{C_2 s}\right) I_2(s) = 0$$

$$\frac{I_1(s)}{E_i(s)} =$$

$$\det \begin{bmatrix} 1 & -\frac{1}{C_1 s} \\ 0 & \frac{1}{C_1 s} + R_2 + \frac{1}{C_2 s} \end{bmatrix}$$

$$\det \begin{bmatrix} R_1 + \frac{1}{C_1 s} & -\frac{1}{C_1 s} \\ -\frac{1}{C_1 s} & \frac{1}{C_1 s} + R_2 + \frac{1}{C_2 s} \end{bmatrix}$$

$$\det \begin{bmatrix} R_1 + \frac{1}{C_1 s} & 1 \\ -\frac{1}{C_1 s} & 0 \end{bmatrix}$$

$$\frac{I_2(s)}{E_i(s)} =$$

$$\det \begin{bmatrix} R_1 + \frac{1}{C_1 s} & -\frac{1}{C_1 s} \\ -\frac{1}{C_1 s} & \frac{1}{C_1 s} + R_2 + \frac{1}{C_2 s} \end{bmatrix}$$

$$= \frac{\frac{1}{C_1 s}}{(R_1 + \frac{1}{C_1 s})(\frac{1}{C_1 s} + R_2 + \frac{1}{C_2 s}) - \frac{1}{C_1 s^2}}$$

$$= \frac{\frac{1}{C_1 s}}{\frac{R_1}{C_1 s} + R_1 R_2 + \frac{R_1}{C_2 s} + \cancel{\frac{1}{C_1 s}} + \frac{R_2}{C_1 s} + \frac{1}{C_1 C_2 s} - \cancel{\frac{1}{C_1 s}}}$$

$$= \frac{\frac{1}{C_1 s}}{\frac{R_1 C_2 s + R_1 C_1 R_2 C_2 s + R_1 C_1 s + R_2 C_2 s + 1}{C_1 C_2 s}}$$

$$G(s) = \frac{C_2 s}{R_1 C_1 R_2 C_2 s + (R_1 C_2 + R_1 C_1 + R_2 C_2) s + 1}$$

